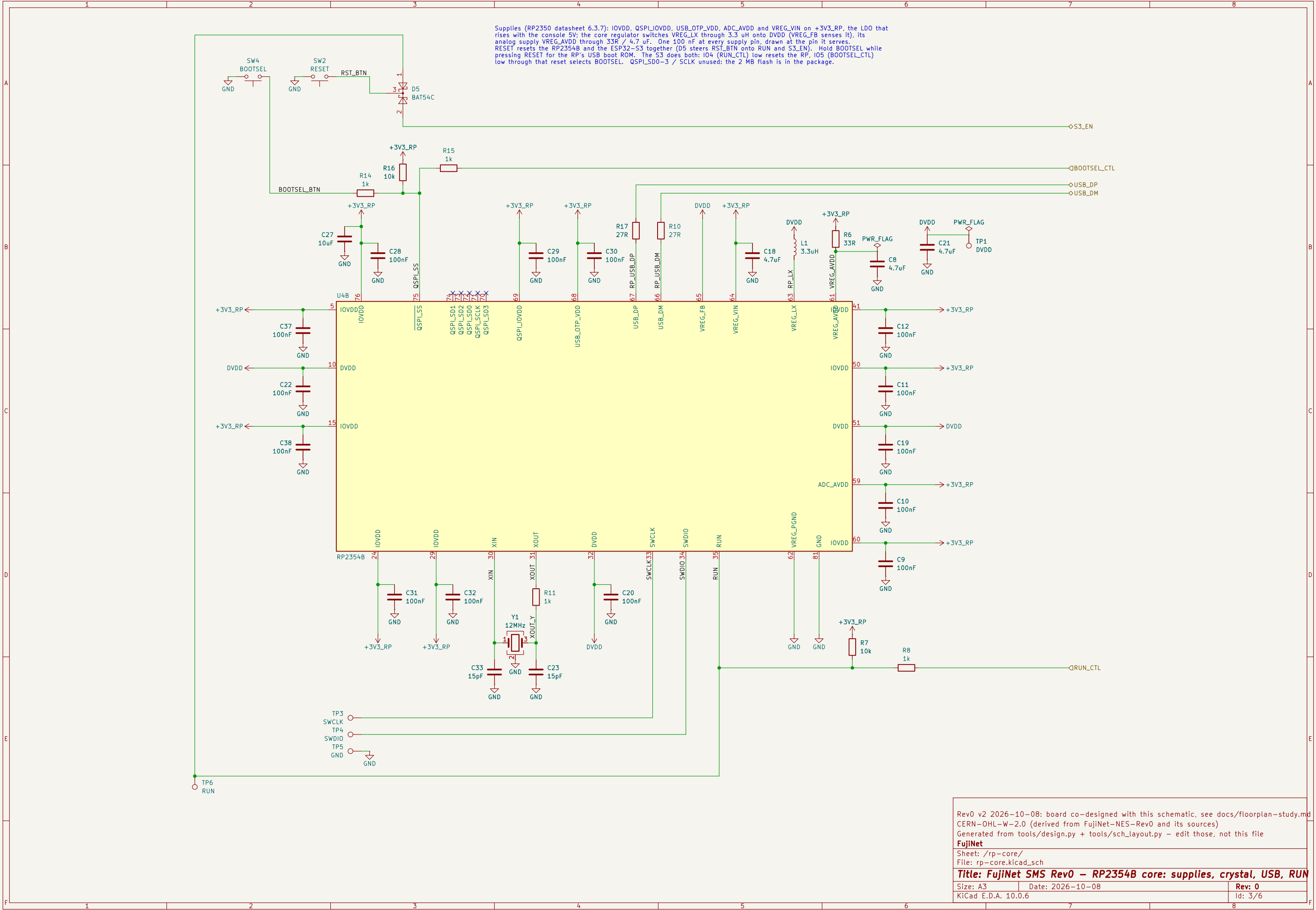
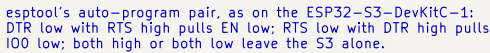
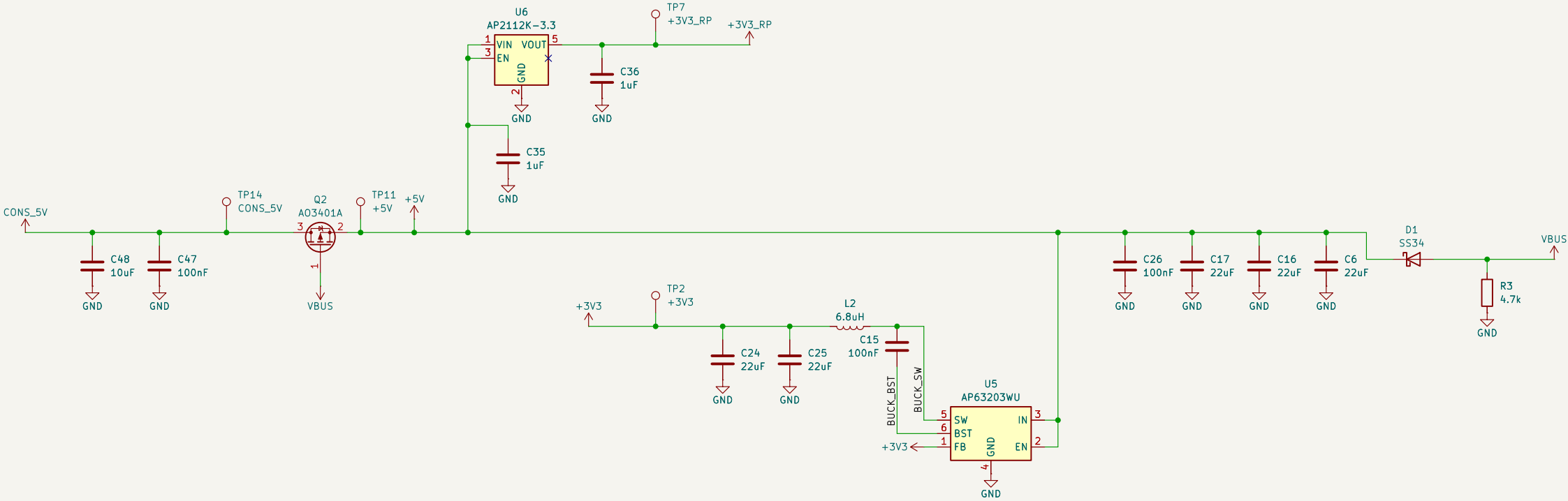






Rev0 v2 2026-10-08: board co-designed with this schematic, see docs/floorplan-study.md
CERN-OHL-W-2.0 (derived from FujiNet-NES-Rev0 and its sources)
Generated from tools/design.py + tools/sch_layout.py - edit those, not this file
FujiNet
Sheet: /usb/
File: usb.kicad_sch
Title: FujiNet SMS Rev0 – USB-C and the CP2102N bridge
Size: A3 Date: 2026-10-08 **Rev: 0**
KiCad E.D.A. 10.0.6 Id: 5/6

Power tree. CONS_5V is the console's 7805 through the edge fingers: about 480 mA peak (the S3 transmitting through the buck), 180 mA average. Q2 A03401A (D = CONS_5V, S = +5V, G = VBUS) is fully on without USB, so the 74HCT glue sees VCC >= 4.5 V (a Schottky drop here would leave no margin): with USB in, its gate goes high and only its body diode conducts, so nothing back-feeds the console. D1 SS34 puts USB VBUS onto +5V. It is reverse-biased whenever the console powers the cart, and its leakage (0.5 mA at 25 C, 20 mA at 100 C) flows into VBUS, the FET gate: R3 4.7k holds VGS at -2.8 V up to 500 uA (tools/audit/spice_checks.py). +5V feeds the 5V glue and both SRAMs (cart-bus), the AP2112K LDO -> +3V3_RP (every RP2354B supply pin: it rises with the console rail, so the 5V-tolerant pads are never unpowered with 5V on them) and the AP63203 buck -> +3V3 (ESP32-S3, microSD, CP2102N). TP14, TP11, TP2, TP7 and the rp-core DVDD pad are bring-up test points.



PWR_FLAG: these rails reach their loads only through passive parts (the edge fingers, USB-C, the P-FET, the SS34, the buck's inductor).

